

Writing Two-Step Equations from Contexts

MA.7.AR.2.2

I can turn a situation into the equation $px + q = r$, and handle a percentage in it.

You can already solve these.

Today the hard part is writing them down.

Warm-Up

On your own. Four minutes.

Same context, different question

A tank holds 2,000 gallons and drains 60 gallons an hour.

- 1.** How much is left after 5 hours?
- 2.** After how many hours are 1,400 gallons left?
- 3.** Write a sentence saying exactly what your letter stands for.

Notes

I. Define the letter in a full sentence

- A. **The unknown is the one thing the question asks for. Give it a letter.**
- B. **t = the number of hours the tank has been draining.**
- C. “ t = hours” is not a definition. t is a number, not a unit.

Notes

II. In $px + q = r$, p is a rate and q happens once

- A. p is almost always a **RATE** — per hour, per item, per month.
- B. q is almost always a **STARTING VALUE** — charged once, no matter what.
- C. r is the total the problem reports at the end.
- D. **If the rate subtracts, p is negative: $2000 - 60t = 1400$.**

Notes

III. A percentage becomes a decimal first

- A. commission — a percent of sales, paid on top of a salary
- B. discount — a percent taken off a price
- C. **Write the percent as a decimal before anything else. $10\% = 0.10$.
 $3\% = 0.03$.**

Notes

IV. “20% off” means you pay 80%

- A. **20% OFF means multiply by 0.80, not by 0.20.**
- B. 0.20p is the discount itself. 0.80p is what you hand over.
- C. **15% ADDED means multiply by 1.15, for the same reason.**

Example 1

A tower is draining.

It starts with 12,340 gallons and loses 780 gallons an hour.

After how many hours are 5,125 gallons left?

Define your letter in a sentence. Say it before you write anything.

Example 1

Write it, then solve it.

$$12,340 - 780t = 5,125$$

$$-780t = -7,215$$

$$t = 9.25$$

$$\text{Check: } 12,340 - 780(9.25) = 12,340 - 7,215 = 5,125 \quad \checkmark$$

Example 1

Say what it means.

After 9.25 hours — 9 hours and 15 minutes — 5,125 gallons are left.

Example 2

A commission.

A salesperson earns \$600 a week plus 4% of everything they sell.

One week they earned \$982. How much did they sell?

Which number is p , and what does it become?

Example 2

Write it, then solve it.

$$600 + 0.04s = 982$$

$$0.04s = 382$$

$$s = 9,550$$

$$\text{Check: } 600 + 0.04(9,550) = 600 + 382 = 982 \quad \checkmark$$

They sold \$9,550 worth.

Example 2

On, and off.

4% ADDED to a price p	4% OFF a price p
you pay $1.04p$	you pay $0.96p$
the extra is $0.04p$	the discount is $0.04p$

Example 3

Victoria's week. Three numbers are in it.

Victoria earns \$350 a week plus a 10% commission on her sales.

She needs her total earnings to be \$800. How much must she sell?

Example 3

Solved. Now read the question again.

$350 + 0.10s = 800$

$0.10s = 450$

$s = 4,500$

Quantity	Value	Where it is in the problem
Her sales	\$4,500	the unknown, s
Her commission	\$450	10% of her sales
Her total earnings	\$800	\$350 salary + \$450 commission

Example 3

Which of the three did the question want?

She must sell \$4,500.

- A. Solving for s is the easy half. Reading which of the three the question wanted is the other half.
- B. **This is a reading error, not an arithmetic error — and it is the one that costs points.**

Whiteboards • Question 1 of 9

Boards up on three.

A gym: \$75 to join, then \$18 a month. Cost after m months?

Write it.

An expression, an equation or a number — read which is being asked for.

Whiteboards · Question 1 of 9

Answer.

A gym: \$75 to join, then \$18 a month. Cost after m months?

the 75 happens once; the 18 happens every month

Answer: $75 + 18m$

Whiteboards • Question 2 of 9

Boards up on three.

A gym: \$75 to join, \$18 a month. The total came to \$273. Write the equation.

Write it.

An expression, an equation or a number — read which is being asked for.

Whiteboards • Question 2 of 9

Answer.

A gym: \$75 to join, \$18 a month. The total came to \$273. Write the equation.

the total goes on the other side

Answer: $75 + 18m = 273$

Whiteboards • Question 3 of 9

Boards up on three.

$75 + 18m = 273$. How many months?

Write it.

An expression, an equation or a number — read which is being asked for.

Whiteboards • Question 3 of 9

Answer.

$75 + 18m = 273$. How many months?

$$18m = 198$$

Answer: $m = 11$ months

Whiteboards • Question 4 of 9

Boards up on three.

A gym with NO joining fee, \$18 a month, \$273 in total. How many months?

Write it.

An expression, an equation or a number — read which is being asked for.

Whiteboards • Question 4 of 9

Answer.

A gym with NO joining fee, \$18 a month, \$273 in total. How many months?

$$18m = 273$$

Answer: $m \approx 15.17$ months

Whiteboards • Question 5 of 9

Boards up on three.

A 15% fee is added to sales of s dollars. What do you pay?

Write it.

An expression, an equation or a number — read which is being asked for.

Whiteboards • Question 5 of 9

Answer.

A 15% fee is added to sales of s dollars. What do you pay?

15% = 0.15, and you pay the sales AND the fee

Answer: $1.15s$

Whiteboards • Question 6 of 9

Boards up on three.

A jacket priced p dollars is 25% off. What do you pay?

Write it.

An expression, an equation or a number — read which is being asked for.

Whiteboards • Question 6 of 9

Answer.

A jacket priced p dollars is 25% off. What do you pay?

25% off means you keep 75%

Answer: $0.75p$

Whiteboards • Question 7 of 9

Boards up on three.

A giraffe is 6 feet tall at birth and grows 0.08 foot a day. It is now 6.48 feet. How many days old?

Write the equation and solve it.

An expression, an equation or a number — read which is being asked for.

Whiteboards • Question 7 of 9

Answer.

A giraffe is 6 feet tall at birth and grows 0.08 foot a day. It is now 6.48 feet. How many days old?

$$6 + 0.08d = 6.48, \text{ so } 0.08d = 0.48$$

Answer: $d = 6$ days

Whiteboards • Question 8 of 9

Boards up on three.

Mr Humphries has some quarters and four dimes, worth \$2.40 altogether. How many quarters?

Write the equation and solve it.

An expression, an equation or a number — read which is being asked for.

Whiteboards • Question 8 of 9

Answer.

Mr Humphries has some quarters and four dimes, worth \$2.40 altogether. How many quarters?

four dimes is \$0.40, so $0.25q + 0.40 = 2.40$

Answer: $q = 8$ quarters

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Shoes are marked 20% off. Maddox also has a \$10 coupon. The store takes the 20% off FIRST, then the coupon. He paid \$36.20. What was the original price?

Write your answer in sentences.

This one is written work. Write the equation first, then solve it.

Whiteboards • Question 9 of 9

Answer.

Shoes are marked 20% off. Maddox also has a \$10 coupon. The store takes the 20% off FIRST, then the coupon. He paid \$36.20. What was the original price?

$$0.80p - 10 = 36.20, \text{ so } 0.80p = 46.20$$

Answer: \$57.75

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.